Certified AC Robust Secure Sets with GENCO in the Loop

GENCO Challenge 2026 entry — **Muhammad Abdillah**, Electrical Engineering Study Program, Universitas Pertamina, Jakarta, Indonesia

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The robust secure set (RSS) is the hourly range of dispatch that stays secure for *every* PV/load realisation in a forecast box, in the base case and in N-1 contingencies (IEEE 118-bus, PGLib, 20 PV buses). This Space shows how a grid foundation model can be used inside a **certified** security workflow:

	Result (all from `results/`)
Two-phase Benders (DC)	48/48 problems certified optimal; big-M MILP misreports 20, by up to 6.3 MW
Exact AC check of DC limits	45/48 certified DC limits are AC-insecure
GENCO-in-the-loop AC correction	secure range 644 → 414 MW on average, empty in hour 18; 1 missed insecure scenario in a full audit of 19,500
Physics-trajectory recalibration	GENCO's own per-layer AC mismatch removes 56 % of its error
Conformal screening, α = 0.05	PTR clears 54 % of secure scenarios (plain: 4.7 %), observed miss 0.09 % ≤ 5 %; unseen topology: 56 %
Second-life EV batteries	+562 MW certified DC range on average

Tabs

1. **Results** — the 24-hour study (DC Benders vs big-M MILP, AC-corrected RSS, screening, batteries).
2. **Live DC-RSS (Benders)** — solve either RSS limit for any hour in about a second and see the certificate.
3. **GENCO probe** — the AC mismatch inside GENCO, layer by layer, for seen and unseen topologies; what the recalibrator learns from it.
4. **Conformal screen (live)** — pick α and the method (PTR / residual-normalised / plain), fit the screen and see which scenarios are cleared without exact AC.

Built with

- **gridfm-datakit 1.1.0** (PF mode, lightsim2grid) — all training data and every exact AC check.
- **gridfm-graphkit 0.9.0.3** — GENCO (`GNS_heterogeneous`, PowerFlow task) training, inference, branch flows; forward hook on the residual layer for the

trajectory probe; autograd adversary.
- scipy/HiGHS for Benders and the big-M baseline, scikit-learn for PTR, Gradio.

Reproduce

```
        bash setup.sh
        python genco_ac_rssr_challenge.py          # data, train, E1, E3, E4,
E5, figures, numbers, docs (~6 h on 2 CPU cores)
        python genco_ac_rssr_challenge.py --steps e1,e3,e4,e5,fig,numbers,docs
# with the shipped weights (~45 min)
        python app.py
```

GENCO used here: 1,077,891 parameters, trained from scratch on CPU because the official checkpoints could not be downloaded from the author's environment; the weights are in `weights/genco_pf_case118/`. To use an official checkpoint: `python genco_ac_rssr_challenge.py --steps e3,e4,fig,numbers,docs --genco-run <dir> --genco-config <yaml>`.

Limitations

No speed claim on this 118-bus CPU case (exact AC 0.29 ms vs GENCO 10.3 ms per scenario): the contribution is the share of exact checks that can be skipped *with a guarantee*. AC-corrected limits are verified on the screened pool, not over the continuous box; generator Q-limits are reported, not enforced; the conformal guarantee assumes exchangeability (the unseen-topology result is empirical).

Paper (with the complete model and all proofs in the appendix): `paper/main.pdf` · Supplement (formulation and proofs): `paper/supplement.pdf` · Post: `docs/submission_post.md` · Numerical checks of every proved statement: `pytest -q tests`